An alternative to generating completely random keys during manufacturing is to derive device-specific keys from a master key and some device-specific information. For example, you can use HKDF from chapter 11 to derive a unique device-specific key based on a unique serial number or ethernet MAC address assigned to each device. The derived key is stored on the device as before, but the API server can derive the key for each device without needing to store them all in a database. When the device connects to the server, it authenticates by sending the unique information (along with a timestamp or a random challenge to prevent replay), using its device key to create a MAC. The server can then derive the same device key from the master key and use this to verify the MAC. For example, Microsoft's Azure IoT Hub Device Provisioning Service uses a scheme similar to this for group enrollment of devices using a symmetric key, see https://docs.microsoft.com/en-us/azure/iot-dps/concepts-symmetric-key-attestation.

12.4.2 Key distribution servers

Rather than installing a single key once when a device is first acquired, you can instead periodically distribute keys to devices using a key distribution server. In this model, the device uses its initial key to enroll with the key distribution server and then is supplied with a new key that it can use for future communications. The key distribution server can also make this key available to API servers when they need to communicate with that device.

> **LEARN MORE** The E4 product from Teserakt (https://teserakt.io/e4/) includes a key distribution